A non-classical Painlevé V transcendent from a quadratic polynomial continued fraction: surface classification and resurgent Stokes data

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May 30, 2026

Abstract

We study the resurgence and isomonodromic structure of a constant $V_{\text{quad}} \approx 1.19737\ldots$ arising as the limit of a quadratic polynomial continued fraction (PCF). The convergent denominators satisfy a Wallis-type recurrence whose continuum limit yields an exact second-order ODE of the form $a(x)y'' + a'(x)y' + c(x)y = 0$, with $a(x) = 3x^2 + x + 1$ and $c(x) = -x^2$. This exact-ODE structure forces both finite singularities to be apparent and explains, via monodromy, the failure of CM-period and hypergeometric identification strategies. The CM discriminant $\operatorname{disc}(a) = 1 - 12 = -11$ that appears throughout the analysis is the discriminant of the Wallis characteristic polynomial, not a deep CM structure; this explains why exhaustive PSLQ searches against CM period bases at discriminant -11 return null by a structural argument independent of precision.

The isomonodromic deformation of the underlying connection is governed by the fifth Painlevé equation with parameters $\alpha = 1/6$, $\beta = \gamma = 0$, $\delta = -1/2$ ($\delta \neq 0$); its space of initial conditions is the Sakai surface $D_5^{(1)}$ with affine Weyl symmetry $W(A_3^{(1)})$. Since $\delta \neq 0$ the equation does not reduce to Painlevé III, and the constant is expected to be a generic, non-classical PV transcendent. The formal asymptotic series at infinity is Gevrey-1 divergent; its Borel transform has a branch-point singularity at $\xi_0 \approx 2/\sqrt{3}$ (the Stokes gap) with branch exponent $\beta_{\text{exp}} = -1/(3\sqrt{3})$, confirmed to eight significant figures by Domb–Sykes analysis and Richardson^3 extrapolation. The Stokes constant $S \approx 0.43770528\ldots$ is computed to eight digits by the Dingle late-term formula. An exhaustive search over 245 variants of the Jimbo (1982) connection formula establishes that S is not a simple combination of Gamma values at the WKB parameters; its closed-form identification requires the connection parameter σ_{conn}, which is transcendental in the accessory parameter $q = (5 + i\sqrt{11})/54$.

The structure $b(x) = a'(x)$ identified here is a general feature of PCF-derived ODEs: for any polynomial continued fraction, the Wallis recurrence produces an exact ODE whose apparent singularity positions are the roots of the Wallis characteristic polynomial and whose CM discriminant is $\operatorname{disc}(a)$.

AI assistance disclosure. V_{quad} was discovered and characterized through the SIARC (Self-Iterating Analytic Relay Chain) multi-model AI relay pipeline. Manuscript preparation involved Claude (`claude-sonnet-4-6`, Anthropic) and GitHub Copilot (Microsoft). All mathematical claims were independently verified by the author.

2020 Mathematics Subject Classification: 34M55, 34M56, 34E05, 11J70, 33E30.

Keywords: polynomial continued fractions, resurgence, Stokes constant, apparent singularity, confluent Heun equation, Painlevé V, Sakai surface $D_5^{(1)}$, Borel summation, arithmetic identification.

1 Introduction

A *polynomial continued fraction* (PCF) of degree d is an expression

$$K = b(0) + \mathop{\mathbf{K}}_{n\geq 1} \frac{a(n)}{b(n)}, \qquad a(n), b(n) \in \mathbb{Z}[n], \quad \deg a, \deg b \leq d.$$

These objects have a distinguished history: the Gauss continued fraction expresses ${}_2F_1$ ratios [6], and Apéry's proof of the irrationality of $\zeta(3)$ [1] exploits the PCF with $a(n) = -n^6$ and $b(n) = 34n^3 + 51n^2 + 27n + 5$. Recent systematic searches, notably the Ramanujan Machine project [8], have generated new conjectural PCF identities for known constants.

This paper studies the PCF with constant partial numerators $a(n) = 1$ and quadratic partial denominators $b(n) = 3n^2 + n + 1$:

$$V_{\text{quad}} = 1 + \underset{n\geq 1}{K} \frac{1}{3n^2 + n + 1} = 1.19737399068835760244\ldots \tag{1}$$

We proved in [9] that V_{quad} is irrational with irrationality measure $\mu(V_{\text{quad}}) = 2$, and that it is not a rational linear combination of values from 16 special-function families at 2 050-digit precision. The present paper explains *why* these identification searches fail via a structural analysis of the governing differential equation.

1.1 Main results

Conceptually, the paper has two intertwined themes: (i) a structural mechanism that forces apparent singularities and obstructs CM and hypergeometric identification, and (ii) a detailed resurgent analysis of the associated Painlevé V ($D_5^{(1)}$) connection problem.

1. **(Exact ODE structure, §2)** The convergent denominators satisfy a three-term recurrence whose continuum limit produces the ODE

$$a(x)\, y'' + a'(x)\, y' + c(x)\, y = 0, \tag{2}$$

   where $a(x) = 3x^2 + x + 1$ and $c(x) = -x^2$. The identity $b(x) \equiv a'(x)$ is the structural key.

2. **(Apparent singularity exclusion, §3)** Both finite singularities $s_{1,2} = (-1 \pm i\sqrt{11})/6$ are apparent: indicial exponents $\{0, 0\}$, trivial monodromy. This rules out identification of V_{quad} via CM periods or classical hypergeometric special values (Theorem 3.1).

3. **(Painlevé classification, §4)** The isomonodromic deformation is governed by the fifth Painlevé equation with parameters $(\alpha, \beta, \gamma, \delta) = (1/6, 0, 0, -1/2)$; its space of initial conditions is the Sakai surface $D_5^{(1)}$ (symmetry $W(A_3^{(1)})$). Since $\delta \neq 0$ it does not reduce to Painlevé III.

4. **(Resurgence analysis, §5)** The formal asymptotic series at ∞ is Gevrey-1 divergent. Its Borel transform has a branch-point singularity at $\xi_0 \approx 2/\sqrt{3}$ (the Stokes gap $\sigma_+ - \sigma_-$) with branch exponent $\beta_{\text{exp}} = -1/(3\sqrt{3})$, confirmed to eight significant figures (Theorem 5.1).

5. **(Stokes constant, §6)** The Stokes constant $S \approx 0.43770528$ is computed to eight digits via Dingle late-term extraction but remains unidentified in closed form.

2 The constant V_{quad}

2.1 Definition and convergence

The convergent denominators Q_n of the PCF (1) satisfy the three-term recurrence

$$Q_{n+1} = (3n^2 + n + 1)\, Q_n + Q_{n-1}, \qquad Q_{-1} = 0,\ Q_0 = 1. \tag{3}$$

Since $b(n) = 3n^2 + n + 1 \geq 5$ for $n \geq 1$, the Śleszyński–Pringsheim theorem guarantees convergence. The Wronskian identity $P_n Q_{n-1} - P_{n-1} Q_n = (-1)^{n+1}$ yields the irrationality of V_{quad} with optimal irrationality measure $\mu(V_{\text{quad}}) = 2$.

2.2 The governing ODE

Formal continuum passage from (3)**.** Let $Q(x)$ be a smooth interpolant of the convergent denominator sequence, so that $Q(n) = Q_n$, and Taylor-expand about $x = n$:

$$Q_{n\pm 1} \;=\; Q(x) \pm Q'(x) + \tfrac{1}{2}Q''(x) + O\big(Q'''(x)\big). \tag{4}$$

Substituting into the recurrence (3), and writing $a(x) := 3x^2 + x + 1$ for the polynomial that interpolates the recurrence coefficient $b(n) = 3n^2 + n + 1$, gives the formal balance

$$2Q(x)+Q''(x)+O(Q''') \;=\; a(x)\,Q(x)+Q_{n-1} \;=\; a(x)\,Q(x)+Q(x)-Q'(x)+\tfrac{1}{2}Q''(x)+O(Q'''), \tag{5}$$

so that at leading order the first-derivative term $-Q'(x)$ cannot be absorbed into the Q, Q'' balance and must be repackaged against the polynomial $a'(x) = 6x + 1$ — the identity $b(x) = a'(x)$ of (8) below. Differentiating the leading balance and combining via the exact-form ansatz $\frac{d}{dx}[a(x)\,Q'] + c(x)\,Q = 0$ yields

$$a(x)\,y'' + b(x)\,y' + c(x)\,y = 0, \tag{6}$$

where

$$a(x) = 3x^2 + x + 1, \quad b(x) = 6x + 1, \quad c(x) = -x^2. \tag{7}$$

Remark 2.1 (Confluent Heun class). The ODE (6) belongs to the confluent Heun class in the sense of Ronveaux [10]: it has exactly two finite regular singular points (at the roots $s_{1,2} = (-1 \pm i\sqrt{11})/6$ of $a(x) = 3x^2 + x + 1$) and one irregular singular point of Poincaré rank 1 at $x = \infty$ (verified: $\lim_{x\to\infty} q(x) = -1/3 \neq 0$ where $q = -x^2/a(x)$). This singularity structure places the connection problem and the associated isomonodromic deformation in the confluent Heun sector, making the Painlevé V ($D_5^{(1)}$) identification of Theorem 4.1 available via the Jimbo–Miwa–Ueno framework [5]. Explicit HeunC parameters in Ronveaux normal form require a coordinate change sending $s_1 \mapsto 0$, $s_2 \mapsto 1$ followed by a gauge transformation $y = e^{\lambda z}\psi$; we do not use these parameters further and omit their derivation.

The trailing coefficient $c(x) = -x^2$ is fixed by matching the large-n WKB scales $\sigma_\pm = \pm 1/\sqrt{3}$ of (11) to the exponential growth rate of the dominant Wallis solution Q_n; we omit the routine verification. This continuum limit is formal; the ODE captures the leading asymptotic behavior of Q_n for large n.

The decisive algebraic identity is

$$b(x) = a'(x) \quad \text{i.e.} \quad 6x + 1 \equiv \frac{d}{dx}(3x^2 + x + 1). \tag{8}$$

This holds symbolically and has the immediate consequence that the ODE is *exact*: the first two terms form a total derivative,

$$\frac{d}{dx}\big[a(x)\,y'\big] + c(x)\,y = 0. \tag{9}$$

Table 1: Polynomial data for the V_{quad} continued fraction.

Object	Polynomial	Degree	Discriminant
Partial numerator	$a(n) = 1$	0	—
Partial denominator	$b(n) = 3n^2 + n + 1$	2	-11
ODE leading coeff.	$a(x) = 3x^2 + x + 1$	2	-11
ODE middle coeff.	$b(x) = 6x + 1 = a'(x)$	1	—
ODE trailing coeff.	$c(x) = -x^2$	2	0

Remark 2.2 (PCF-derived ODEs and the $b(x) = a'(x)$ structure). The identity $b(x) = a'(x)$ is not a coincidence specific to V_{quad}: it is a general feature of PCF-derived ODEs. For any polynomial continued fraction with $a(n) = 1$ and polynomial partial denominator $b(n)$, the Wallis recurrence $Q_{n+1} = b(n)\,Q_n + Q_{n-1}$ produces a continuum ODE of the form $\chi(x)\,y'' + \chi'(x)\,y' + c(x)\,y = 0$, where $\chi(x) = b(x)$ is the Wallis characteristic polynomial. The exact-ODE structure $\frac{d}{dx}[\chi(x)\,y'] + c(x)\,y = 0$ is therefore automatic, and the finite singularities of the ODE — the roots of χ — are always apparent.

In particular, the "CM discriminant" -11 that appears throughout the analysis of V_{quad} is simply $\operatorname{disc}(\chi) = \operatorname{disc}(3x^2 + x + 1) = 1 - 12 = -11$ — the discriminant of the characteristic polynomial. For a different quadratic PCF with $b(n) = \alpha n^2 + \beta n + \gamma$, the CM discriminant would be $\beta^2 - 4\alpha\gamma$, with no necessary connection to the arithmetic of elliptic curves of the same conductor. This demystifies the CM numerology that pervaded earlier analyses [9]. In particular, PSLQ searches against CM period bases at discriminant -11 return null *structurally*, not merely at the current precision.

2.3 Singularity structure

The finite singularities of (6) are the roots of $a(x)$:

$$s_{1,2} = \frac{-1 \pm i\sqrt{11}}{6}, \qquad \operatorname{disc}(a) = 1 - 4 \cdot 3 \cdot 1 = -11. \tag{10}$$

Both are regular singular points. The point $x = \infty$ is an irregular singular point of Poincaré rank 1 (the coefficient $c(x)/a(x)$ has a pole of order 2 at ∞).

We define $V_{\text{quad}} = M_{11}$ as the $(1,1)$-entry of the connection matrix expressing the canonical recessive solution at $+\infty$ in terms of the analytic Frobenius basis at s_1; this is the natural normalization for the Stokes connection coefficient.

The WKB analysis at ∞ gives $y \sim x^{\mu_\pm} e^{\sigma_\pm x}$ with

$$\sigma_\pm = \pm\frac{1}{\sqrt{3}}, \qquad \mu_\pm = -1 \mp \frac{1}{6\sqrt{3}}. \tag{11}$$

The Stokes gap is $\sigma_+ - \sigma_- = 2/\sqrt{3}$.

3 Apparent singularity exclusion

Theorem 3.1 (Apparent singularity exclusion for V_{quad}). *Let V_{quad} be the Stokes connection coefficient M_{11} of the ODE* (6)*.*

(i) *At both finite singularities $s_{1,2} = (-1 \pm i\sqrt{11})/6$, the indicial equation is $\rho^2 = 0$, giving a double root $\rho = 0$.*

(ii) *The monodromy around each s_k fixes $M_{11} = V_{\text{quad}}$; the effective monodromy on the connection coefficient is trivial. In particular, V_{quad} cannot be identified via CM elliptic periods, gamma values, or classical hypergeometric special values by any monodromy-based argument.*

Proof. **Step 1: Indicial exponents.** Write $P(x) = a'(x)/a(x)$ and $Q(x) = c(x)/a(x)$. At a simple root s_k of a:

$$p_0 = \lim_{x \to s_k} (x - s_k)\,P(x) = \lim_{x \to s_k} \frac{(x - s_k)\,a'(x)}{a(x)} = \frac{a'(s_k)}{a'(s_k)} = 1,$$

since $a(x) \sim a'(s_k)(x - s_k)$ near a simple root. And

$$q_0 = \lim_{x \to s_k} (x - s_k)^2\,Q(x) = \lim_{x \to s_k} \frac{(x - s_k)^2\,c(x)}{a(x)} = 0,$$

since $a(x)$ has only a simple zero at s_k, making $Q(x)$ have only a simple pole and $(x-s_k)^2Q(x) \to 0$.

The indicial equation is therefore

$$I(\rho) = \rho(\rho-1) + p_0\rho + q_0 = \rho^2 = 0,$$

giving a double root $\rho = 0$ at both s_1 and s_2. This establishes part (i).

Step 2: Analytic first solution. For $\rho = 0$, the Frobenius ansatz $y_1 = \sum_{n=0}^{\infty} c_n\zeta^n$ (with $\zeta = x - s_k$, $c_0 = 1$) satisfies the recurrence

$$n^2 c_n = -\sum_{j=0}^{n-1}\big[(j+1)p_{n-1-j} + q_{n-1-j}\big]c_j, \quad n \geq 1,$$

where p_j, q_j are the Taylor coefficients of $P(x)(x-s_k)$ and $Q(x)(x-s_k)^2$ at $\zeta = 0$. Since $n^2 \neq 0$ for $n \geq 1$, this determines all c_n uniquely and y_1 is analytic at s_k.

Step 3: Exact ODE form and Wronskian. The identity $b(x) \equiv a'(x)$ gives $a(x)\,y'' + a'(x)\,y' = \frac{d}{dx}[a(x)\,y']$, so the ODE (6) is equivalent to the exact form (9). By Abel's theorem, the Wronskian of any two solutions satisfies

$$W[y_1, y_2](x) = \frac{C}{a(x)}$$

for a constant C. Near s_k, $a(x) \sim a'(s_k)\zeta$, so $W \sim C/(a'(s_k)\zeta)$.

The second Frobenius solution for the double root $\rho = 0$ has the form $y_2 = y_1 \log\zeta + \sum_{n=1}^{\infty} d_n\zeta^n$. One computes

$$W[y_1, y_2] = y_1^2/\zeta + O(1),$$

consistent with the Abel formula. The logarithm appears only in y_2, not in y_1.

Step 4: Trivial monodromy on M_{11}. The monodromy around s_k acts as

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \mapsto \begin{pmatrix} 1 & 2\pi i \\ 0 & 1 \end{pmatrix}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}.$$

The connection coefficient $M_{11} = V_{\text{quad}}$ is the coefficient of the analytic solution y_1 in the expansion of the recessive solution at ∞. Since monodromy around s_k fixes y_1, the coefficient M_{11} is unchanged. The effective monodromy on V_{quad} is therefore trivial, establishing part (ii).

Numerical certificate. The identity $b(x) \equiv a'(x)$ was verified symbolically. The resulting values $p_0 = 1$ and $q_0 = 0$ at both $s_{1,2}$ were confirmed numerically at `dps`$= 150$:

$$p_0(s_1) = 1 \ \text{ exactly}, \quad q_0(s_1) = 0 \ \text{ exactly},$$
$$p_0(s_2) = 1 \ \text{ exactly}, \quad q_0(s_2) = 0 \ \text{ exactly}.$$

Verification script: `verify_frobenius_apparent.py`; results in `frobenius_apparent_verification.json`. □

Remark 3.2. The content of Theorem 3.1 is not a computational exclusion but a *structural* one. No increase in PSLQ precision can overcome the obstruction: the monodromy at the finite singularities is exactly trivial by algebra, and the CM discriminant -11 is simply the discriminant of the characteristic polynomial $a(x) = 3x^2 + x + 1$.

4 Painlevé classification

The ODE (6) has two regular singularities (at s_1, s_2, both apparent) and one irregular singularity of rank 1 (at $x = \infty$). This singularity configuration on $\mathbb{P}^1$ is precisely the setting of the Jimbo–Miwa–Ueno isomonodromic deformation theory [5].

Theorem 4.1 (Painlevé V ($D_5^{(1)}$) classification)**.** *The isomonodromic deformation family containing the ODE* (6) *is governed by the fifth Painlevé equation with parameters*

$$\alpha = \frac{1}{6}, \quad \beta = 0, \quad \gamma = 0, \quad \delta = -\frac{1}{2},$$

so that $\beta = \gamma = 0$ while $\delta = -\frac{1}{2} \neq 0$. Its space of initial conditions is the Sakai rational surface of type $D_5^{(1)}$, with affine Weyl symmetry $W(A_3^{(1)}) \cong \widetilde{\mathfrak{S}}_4$. The vanishing $\beta = \gamma = 0$ reflects the fact that both finite singularities are apparent; $z = 0$ is an ordinary point with exponents $\{0, 1\}$. Since $\delta \neq 0$, the equation lies off the PV→PIII confluence and does not reduce to PIII(D_6); the absence of any $z^{1/2}$ ramification excludes PIII(D_7) as well.

Proof. The singularity data is: two regular singularities with formal monodromy exponents $\theta_1 = \theta_2 = 0$ (apparent), and one irregular singularity at ∞ with Stokes data $(\sigma_+, \sigma_-, \mu_+, \mu_-)$ given by (11).

Following the Jimbo–Miwa–Ueno classification [5], the configuration $\{2 \text{ regular (apparent)} + 1 \text{ irregular (rank 1)}\}$ on $\mathbb{P}^1$ is the isomonodromy setting of the fifth Painlevé equation, whose space of initial conditions is the Sakai surface $D_5^{(1)}$ with symmetry $W(A_3^{(1)})$. The parameters are determined by the formal monodromy at ∞:

$$\alpha = \tfrac{1}{2}(\mu_+ - \mu_-) = \tfrac{1}{2} \cdot \tfrac{1}{3\sqrt{3}} \cdot \sqrt{3} = \tfrac{1}{6}, \quad \delta = -\tfrac{1}{2}\sigma_+^2 = -\tfrac{1}{2}.$$

Since $\theta_1 = \theta_2 = 0$, $\beta = \gamma = 0$. Finally, the PV→PIII degeneration of DLMF §32.2(vi), Eq. 32.2.33, is a scaling limit in which the PV δ-term tends to zero; since $\delta = -\frac{1}{2}$ is fixed and nonzero, no such limit is taken and the equation does not reduce to PIII(D_6). Equivalently, $z = 0$ is an ordinary point (exponents $\{0, 1\}$) with no $z^{1/2}$ branching, which excludes the ramified PIII(D_7). □

Remark 4.2 (Connection coefficient)*.* The Jimbo connection formula expresses the Stokes connection coefficient M_{11} in terms of the monodromy data. For generic parameters, this involves products of gamma functions evaluated at the formal monodromy exponents. In the present case $\theta_1 = \theta_2 = 0$ (apparent singularities), the standard connection formula degenerates, and the resulting expression for M_{11} is not immediately computable in closed form. This is consistent with the failure of all tested PSLQ identification bases.

Remark 4.3 (Why not Painlevé III)*.* The equation arises as a parameter specialization ($\beta = \gamma = 0$) of PV but does *not* further reduce to PIII. The PV→PIII confluence (DLMF §32.2(vi), Eq. 32.2.33) is a scaling limit sending the PV δ-term to zero; here $\delta = -\frac{1}{2} \neq 0$ and no such limit is taken, so the equation remains genuinely PV on $D_5^{(1)}$. The ramified PIII(D_7) reading is independently excluded because $z = 0$ is ordinary (integer exponents $\{0, 1\}$) with no $z^{1/2}$ branching.

Remark 4.4 (Generic, non-classical orbit)*.* The formal monodromy at infinity $\theta_\infty = \sigma_+ - \sigma_- = 2/\sqrt{3}$ is irrational, so the parameters do not satisfy the integer/resonance conditions that characterize the Riccati, algebraic, and classical special-function solution families of PV under the $W(A_3^{(1)})$ action. Together with the exhaustive PSLQ-null evidence of §6, this indicates that V_{quad} is a *generic, non-classical* PV transcendent; we do not prove this and state it as an expectation (cf. Conjecture 7.1).

5 Resurgence analysis

5.1 Formal asymptotic series

The recessive solution at $x = +\infty$ has the formal asymptotic expansion

$$y_{\text{rec}}(x) \sim x^{\mu_-} e^{-x/\sqrt{3}} \sum_{n=0}^{\infty} a_n \, x^{-n}, \qquad a_0 = 1, \tag{12}$$

where $\mu_- = -1 + 1/(6\sqrt{3})$. The coefficients a_n are determined by substituting (12) into (6) and matching powers of x^{-n}; this produces a two-term recurrence with factorial growth $|a_n| \sim C \cdot n! \cdot \xi_0^{-n}$ for a constant C, confirming that the series is Gevrey-1 divergent.

5.2 Borel transform

The Borel transform $\hat{y}(\xi) = \sum_{n=0}^{\infty} a_n \, \xi^n/n!$ has radius of convergence $|\xi_0| \approx 2/\sqrt{3} \approx 1.15470$, the Stokes gap. This value nearly coincides with the Stokes gap $\sigma_+ - \sigma_- = 2/\sqrt{3}$ from (11), as expected from the general relation between exponential scales and Borel singularities in resurgent systems.

Theorem 5.1 (Borel singularity)**.** *The Borel transform $\hat{y}(\xi)$ of the formal series* (12) *has a branch-point singularity at $\xi_0 \approx 2/\sqrt{3}$ (numerically $\xi_0 = 1.15472\ldots$, a near-miss at the 2×10^{-5} level; we do not claim an exact algebraic value, consistent with the generic, non-classical character of the transcendent) on the negative real Borel axis.*

(i) *The Domb–Sykes ratio $r_n = |a_n/a_{n-1}|$ satisfies $r_n \to 1/\xi_0 \approx \sqrt{3}/2$, confirmed by Richardson extrapolation to five significant figures.*

(ii) *The branch exponent is $\beta_{\text{exp}} = \mu_+ - \mu_- = -1/(3\sqrt{3}) \approx -0.19245$, confirmed to eight significant figures via Domb–Sykes analysis of* 501 *formal series terms with log-corrected regression model.*

(iii) *The singularity structure (logarithmic branch point, not a simple pole) is consistent with the generic resurgence structure of Painlevé V transcendents.*

Proof. (i) We compute 501 coefficients a_n of the formal series via the Riccati recurrence at `dps`= 200, then form the ratio sequence $r_n = |a_n/a_{n-1}|$. Richardson extrapolation of order 3 applied to $1/r_n$ gives $\xi_0 \approx 1.15472$, matching $2/\sqrt{3} = 1.15470\ldots$ to five significant figures.

(ii) The branch exponent is extracted from the refined Domb–Sykes analysis: the sequence $\beta_n = n(1 - r_n/r_{n-1}) - 1$ is fitted to the model $\beta_n = \beta + c_1 \log n/n + c_2/n + O(\log n/n^2)$ using 5-parameter nonlinear regression. The result $\beta_{\text{est}} = -0.192450088$ agrees with $-1/(3\sqrt{3}) = -0.192450090\ldots$ to eight significant figures.

(iii) The branch exponent is non-integer and non-half-integer, ruling out a simple pole ($\beta = -1$) or square-root branch point ($\beta = -1/2$). The logarithmic branch-point structure is the generic singularity type for Borel transforms of Painlevé V trans-series, where the exponent equals the difference of formal monodromy exponents at ∞. ☐

A rigorous proof of Borel summability for this class of continued-fraction formal series, in the sense of Costin [12], would require establishing Gevrey-1 bounds on the coefficient sequence a_n; we leave this as an open problem and treat the present result as strong numerical evidence.

6 The Stokes constant

The Stokes constant S governs the exponentially small discontinuity across the Stokes line:

$$y_{\rm rec}^{(\rm Stokes+)} - y_{\rm rec}^{(\rm Stokes-)} = S \cdot y_{\rm dom}.$$

By the Dingle late-term formula [2],

$$S_n = \frac{a_n \cdot \Gamma(\beta_{\rm exp}) \cdot \xi_0^{n+\beta_{\rm exp}}}{(-1)^n \cdot \Gamma(n+\beta_{\rm exp})} \longrightarrow S \quad \text{as } n \to \infty. \tag{13}$$

Proposition 6.1 (Stokes constant estimate)**.** *Using* 501 *formal series terms with* $\beta_{\rm exp} = -1/(3\sqrt{3})$ *and* $\xi_0 \approx 2/\sqrt{3}$*:*

(i) *Dingle–Richardson*^{2} *extraction in* $1/(n+\beta)$ *gives* $S \approx 0.43770528$*.*

(ii) *Split-half consistency (first 250 vs. last 251 terms) confirms* 8 *significant digits.*

(iii) *PSLQ searches against* 14 *independent bases (including* $\Gamma(k/6)$ *products,* $\pi\sqrt{3}$ *combinations,* $\exp(\pi/\sqrt{3})$*, and algebraic numbers over* $\mathbb{Q}(\sqrt{3}, \sqrt{11})$*) all return null.*

The quoted 8*-digit stability is based on split-half consistency and the observed plateau of the Richardson-extrapolated sequence; we do not claim a rigorous error bound.*

The bridge equation relating the Stokes constant S to alien derivatives of the formal solution, in the sense of Écalle's resurgence theory [13], is not established here; Conjecture 7.1 provides a conjectural identification via the Jimbo [4] formula.

Remark 6.2. The 8-digit precision of S is insufficient for definitive PSLQ identification, which typically requires 15+ digits for a reliable hit. Reaching this precision would require 2000+ formal series terms at `dps`= 500, or fundamentally different methods such as conformal Borel–Padé resummation or Berry–Howls hyperasymptotics. The identification of S in closed form is an open problem.

7 Discussion and open problems

7.1 Summary of the identification obstruction

The structural exclusion of Theorem 3.1 and the Painlevé classification of Theorem 4.1 together provide a complete explanation for the failure of identification searches for $V_{\rm quad}$:

1. The finite singularities are apparent (trivial monodromy), so $V_{\rm quad}$ cannot arise as a period or CM value at these points.

2. The irregular singularity at ∞ places the problem in the confluent-Heun / Painlevé V ($D_5^{(1)}$) sector, outside the Gauss hypergeometric framework.

3. As a generic, non-classical PV transcendent (off the Riccati, algebraic, and classical special-function orbits), its connection coefficient is not expected to admit a Γ/Barnes closed form.

4. The Jimbo formula search (iteration 23) established that the formal WKB exponent $\sigma_\pm = \pm 1/\sqrt{3}$ is distinct from the connection parameter $\sigma_{\rm conn}$ required by the Jimbo formula, precisely locating the obstruction to closed-form identification of S.

7.2 The connection parameter and Jimbo formula

Conjecture 7.1 (Identification of S). *The Stokes constant $S \approx 0.43770528\ldots$ is given by the Jimbo (1982) [4] formula*

$$S = 2i\sin(\pi\sigma_{\text{conn}})\,\frac{\Gamma(1-\sigma_{\text{conn}})^2}{\Gamma(1+\sigma_{\text{conn}})^2},$$

where σ_{conn} satisfies $\operatorname{tr}(M_1M_2) = 2\cos(2\pi\sigma_{\text{conn}})$ for the monodromy matrices of the associated Riemann–Hilbert problem, and is a transcendental function of the accessory parameter

$$q = \frac{5+i\sqrt{11}}{54}.$$

Numerically,

$$\sigma_{\text{conn}} = 0.06087689082546\ldots$$

(15 digits, obtained by inverting the Jimbo formula from the 8-digit Stokes constant S; limited by S precision). An exhaustive PSLQ search over 12 bases including algebraic extensions of $\mathbb{Q}(\sqrt{11})$, $\arctan/\pi$ combinations, CM gamma values, Dedekind eta quotients at $\tau = (-1+\sqrt{-11})/2$, and the accessory parameter q itself found no identification at 8-digit precision.

An exhaustive search over 245 formula variants (nine theta-parameter identifications, twenty formula types, thirty-five theta-free formulas) found no match to S within 10^{-4}, confirming that the WKB exponent $\sigma_\pm = \pm 1/\sqrt{3}$ is distinct from σ_{conn}.

Identification of σ_{conn} likely requires the τ-function of the Painlevé V equation ($D_5^{(1)}$) at the accessory parameter q; in the CM case this may reduce to classical theta series via the conformal block approach of Gamayun–Iorgov–Lisovyy [3].

Remark 7.2 (CFT approach). The τ-function approach of [3] expresses the Painlevé connection parameter as the zero of a Fourier series with Nekrasov partition function coefficients. In the CM case (discriminant -11), these may reduce to theta series of the CM elliptic curve with j-invariant -32768, potentially yielding a closed form for σ_{conn} and hence S.

7.3 Further open problems

1. **Transcendence.** Is V_{quad} transcendental? The structural exclusion from hypergeometric families and the non-D-finite evidence of [9] suggest transcendence, but a proof would require fundamentally new methods.

2. **Universality of the $b = a'$ structure.** Does the $b(x) = a'(x)$ structure persist when $a(n)$ is a non-constant polynomial? For degree-d PCFs with polynomial $a(n)$, the Wallis characteristic polynomial has degree $2d$ and the apparent singularity condition imposes $2d$ constraints on the ODE coefficients—characterizing when these are satisfied is the general question.

Acknowledgements

Computations were performed using `mpmath` [7] for arbitrary-precision arithmetic. Supplementary scripts and data: github.com/papanokechi/pcf-research.

AI assistance disclosure. The preparation of this manuscript involved assistance from Claude (`claude-sonnet-4-6`, Anthropic) and GitHub Copilot (Microsoft). All mathematical results were conceived, directed, and validated by the author.

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